

Gen AI Assignment – 10

Name: PRATHVIRAJ GOWDA

USN: 1CR22CD034

E-mail: prgo22csds@cmrit.ac.in

1. What is Artificial Intelligence?

Artificial Intelligence (AI) is a field of computer science that develops systems capable of performing tasks that normally require human intelligence such as learning, reasoning, and decision making.

2. What is Machine Learning?

Machine Learning is a branch of AI where computers learn from data patterns and improve their performance without explicit programming for every task.

3. What is Deep Learning?

Deep Learning is an advanced machine learning technique that uses multiple layers of neural networks to process complex data and recognize patterns.

4. What is a Neural Network?

A neural network is a computing model inspired by the human brain that uses interconnected nodes to process information and detect patterns.

5. What is Training Data?

Training data is the dataset used to teach an AI model so it can identify patterns and make predictions.

6. What is Overfitting?

Overfitting occurs when a model learns the training data too closely and performs poorly when new data is introduced.

7. What is Underfitting?

Underfitting happens when a model is too simple and fails to capture

the important patterns in the data.

8. What is Supervised Learning?

Supervised learning is a method where models are trained using labeled data that contains both inputs and the correct outputs.

9. What is Unsupervised Learning?

Unsupervised learning is a technique where AI finds patterns or relationships in data that does not have labels.

10. What is Reinforcement Learning?

Reinforcement learning is a training approach where an AI agent learns by interacting with an environment and receiving rewards or penalties.

Generative AI

11. What is Generative AI?

Generative AI refers to AI systems that can produce new content such as text, images, music, or code based on learned patterns.

12. What is an LLM?

A Large Language Model (LLM) is an AI model trained on huge text datasets to understand language and generate natural responses.

13. What is a Prompt?

A prompt is the input instruction given to an AI model that guides it to generate a response.

14. What is Prompt Engineering?

Prompt engineering is the technique of designing clear and effective prompts to obtain better outputs from AI models.

15. What does Temperature mean in AI?

Temperature controls the randomness of AI responses; lower values make responses predictable while higher values allow more creativity.

16. What is Tokenization?

Tokenization is the process of dividing text into smaller pieces called tokens so AI models can analyze and process language.

17. What is Fine-tuning?

Fine-tuning involves retraining a pre-trained model with specific datasets to improve its accuracy for certain tasks.

18. What is AI Hallucination?

AI hallucination occurs when a model generates incorrect or made-up information that appears believable.

19. What is Zero-shot Prompting?

Zero-shot prompting means asking an AI model to perform a task without giving it prior examples.

20. What is Few-shot Prompting?

Few-shot prompting provides a small number of examples to help the AI model understand the task better.

Applications of AI

21. Healthcare

AI assists in healthcare by analyzing medical data, helping doctors diagnose diseases, and improving patient treatment.

22. Finance

In finance, AI is used for fraud detection, financial forecasting, and analyzing market risks.

23. Education

AI supports personalized education by adjusting learning materials based on a student's progress.

24. E-commerce

AI enhances online shopping experiences by recommending products

based on user behavior and preferences.

25. What is a Chatbot?

A chatbot is an AI-based application that interacts with users through text or voice to answer questions or provide services.

Core AI Concepts

26. Natural Language Processing (NLP)

NLP is a field of AI that enables computers to understand, interpret, and generate human language.

27. Computer Vision

Computer Vision allows machines to analyze images or videos and extract useful information from them.

28. Dataset

A dataset is a collection of structured data used to train and test AI models.

29. Model Accuracy

Model accuracy measures how correctly an AI system predicts outcomes compared to total predictions made.

30. AI Bias

AI bias occurs when an AI model produces unfair or inaccurate results due to biased or incomplete training data.

Agentic AI

31. Agentic AI

Agentic AI refers to intelligent systems that can independently plan, make decisions, and take actions to achieve goals.

32. AI Agent

An AI agent is a software system that observes its surroundings and

performs actions to accomplish specific tasks.

33. Memory in AI Agents

Memory allows AI agents to store previous interactions and use that information for better future decisions.

34. Tool Usage

AI systems can integrate with external tools like APIs, databases, or search engines to complete tasks more effectively.

35. Automation in AI

Automation in AI refers to using intelligent systems to perform repetitive or complex tasks with minimal human effort.

Ethics and Future of AI

36. AI Ethics

AI ethics focuses on ensuring AI technologies are developed and used in a responsible and fair manner.

37. Data Privacy

Data privacy ensures that personal or sensitive information is protected from unauthorized access.

38. Explainable AI

Explainable AI aims to make AI decisions understandable so users can know how results were generated.

39. Artificial General Intelligence (AGI)

AGI refers to a future AI system that would have the ability to perform any intellectual task that a human can do.

40. Importance of AI in the Future

AI will be important in the future as it can increase efficiency, support innovation, and help solve complex global problems.